

NICOLETTE MASHABA

Junior Software Engineer | AI & Full-Stack Development

nene171408@gmail.com | +27 63 152 6795 | South Africa | Remote / Hybrid / On-site

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Junior Software Engineer with hands-on experience building and deploying AI-powered and full-stack web applications. Proficient in C#/.NET Core, Python, JavaScript/TypeScript, REST APIs, and cloud deployment on Microsoft Azure (AZ-204 certified). Demonstrated ability to deliver production-ready software in high-stakes environments, including a hackathon-winning AI compliance tool deployed on Azure and featured on an international technology podcast. Skilled in LLM prompt engineering, secure development practices, and translating real-world problems into scalable, maintainable solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C#, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS, Responsive Design

Backend: Node.js (Express), FastAPI, ASP.NET Core, REST API Design

AI & Automation: OpenAI API, LLM Integration, Prompt Engineering, Structured Outputs

Databases: MongoDB Atlas, SQL Server, PostgreSQL, Firebase

Cloud & Deployment: Microsoft Azure (AZ-204), Vercel, Render, Netlify, Docker (fundamentals)

Testing: Manual testing, bug triage, Postman for API testing

Tools & Practices: Git, GitHub, CI/CD concepts, Agile/Scrum, Environment Variables

PROFESSIONAL EXPERIENCE

Technical Trainer (Volunteer) | TechBridle Foundation

2026 – Present

- Mentor learners in web development, JavaScript, React, and backend engineering concepts.
- Lead code review sessions and guide Git workflows, secure project setup, and debugging practices for junior developers.

Software Engineering Intern | Coast IT (Pty) Ltd

2025

- Contributed to feature development within an Agile/Scrum team, participating in sprint planning and daily stand-ups.
- Performed API testing using Postman, resolved bugs, and participated in peer code reviews to improve application stability and code quality.
- Collaborated with senior engineers to implement and test new application modules, gaining practical exposure to production-level codebases.

AI Model Trainer | Outlier AI

2024

- Evaluated AI-generated outputs and improved response quality through structured, documented feedback aligned with model guidelines.
- Conducted prompt testing and model behaviour analysis, identifying patterns that improved response clarity, accuracy, and reliability.

Junior Navigation Officer | South African Navy

2015 – 2021

- Operated in mission-critical environments requiring precision decision-making, high ownership, and strict adherence to protocol — qualities applied directly to software engineering practice.
- Developed strong team coordination and communication skills across multidisciplinary teams under pressure.

PROJECTS

AIMS Compliance Interrogator | *Hackathon Winner* | *Featured on International Tech Podcast*

- Identified inefficiencies in manual compliance review processes and architected an AI-powered tool to analyse structured inputs and automatically flag risks.
- Designed controlled LLM prompt workflows to ensure consistent, auditable, and explainable outputs — directly applicable to regulated AI environments.
- Enforced backend-only AI calls and secure environment variable configuration to handle sensitive compliance data safely.
- Deployed the full-stack application on Microsoft Azure; project won the hackathon and was featured on an Australian technology podcast.

Tech Stack: *Python, FastAPI, OpenAI API, Microsoft Azure, Prompt Engineering*

EmpowaAI | *AI-Powered Career Navigation Platform* | *Startup* | *Active Development*

Co-building an AI-powered platform helping South African youth navigate career paths, optimise job readiness, and access economic opportunities through intelligent systems.

- Designed and developed the full-stack application integrating React (TypeScript), Node.js/Express, and FastAPI-based AI microservices, orchestrating reliable data flow across the entire stack.
- Engineered the Digital Economic Twin concept to model user profiles and generate personalised career insights and income pathway simulations.
- Built AI-powered features including CV analysis, CV revamp, interview coaching, and skill-based opportunity matching — deployed to beta users on Vercel and Render.
- Iterated on product features based on real user feedback, improving AI response relevance through structured prompt engineering and A/B testing of prompt variants.
- Contributed to product architecture decisions in a startup environment, balancing velocity with code quality and maintainability.

Tech Stack: *React, TypeScript, Node.js, Express, FastAPI, MongoDB Atlas, OpenAI API, Tailwind CSS, Vercel, Render*

BookReviewApp | *Full-Stack Web Application*

- Built a full-stack book review platform with user authentication, content moderation, and full CRUD functionality.
- Designed RESTful APIs using ASP.NET Core and Entity Framework Core; implemented database migrations and enforced clean architecture principles for scalability and data integrity.

Tech Stack: *C#, ASP.NET Core, Entity Framework Core, SQL Server, REST APIs*

Interview Replay | *Live Application*

- Developed an AI-driven mock interview tool that delivers structured, actionable feedback on candidate strengths and improvement areas.
- Iteratively improved AI response quality through structured prompt engineering; deployed and maintained production application using clean Git practices and secure environment configuration.

Tech Stack: *JavaScript, OpenAI API, Netlify*

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- J.P. Morgan Software Engineering Virtual Experience

EDUCATION

Occupational Certificate: Software Engineering (NQF Level 6)

CTU Training Solutions, South Africa

2026

Information Technology: Programming Foundations (NQF Level 4)

CTU Training Solutions, South Africa

2024